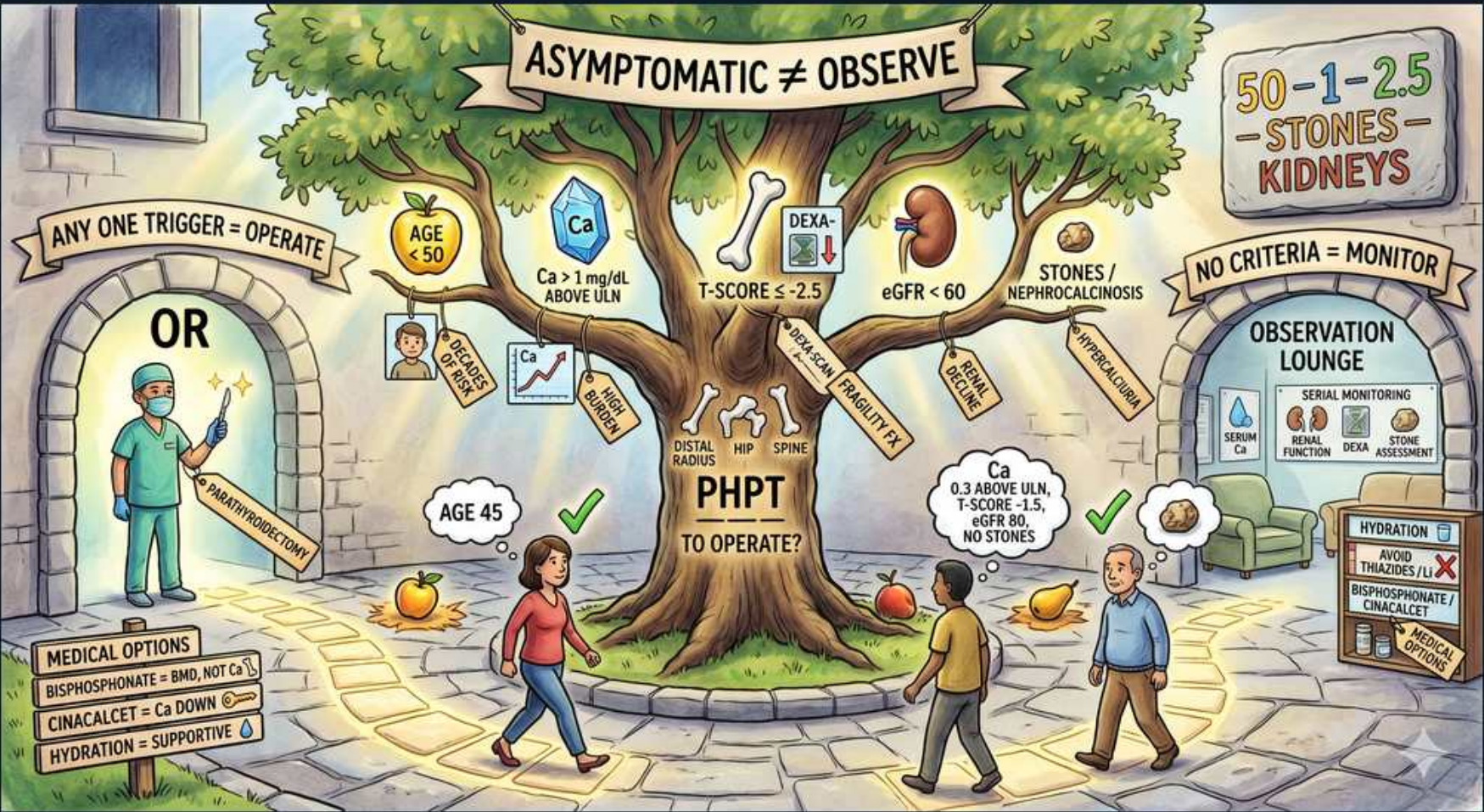


Calcium/PTH/VitD — The PHPT Surgery Decision Tree



1. Asymptomatic ≠ observe banner

WHAT YOU SEE

Top banner: ASYMPTOMATIC ≠ OBSERVE arched over the decision tree — the headline rule that even a symptom-free PHPT patient still needs surgery if any single criterion is met.

WHAT IT ENCODES

Most PHPT today is asymptomatic, found incidentally on a high calcium with an inappropriately normal/high PTH. The central teaching point: asymptomatic does NOT equal observe. Per the 2022 (4th International) Workshop guidelines, meeting ANY single objective criterion mandates referral for parathyroidectomy even without symptoms. Symptomatic disease (overt stones, fractures, neuromuscular/psychiatric symptoms) is always a surgical indication.

BOARD TRAP

A board case shows an asymptomatic patient with PHPT and a T-score of -2.8, then offers 'observe with annual labs since she has no symptoms' — WRONG. Asymptomatic PHPT still requires surgery if any criterion is met; this T-score below -2.5 alone is an operative indication. The discriminator: screen for ALL objective criteria, not just symptoms.

2. '50-1-2.5 stones kidneys' mnemonic

WHAT YOU SEE

Top-right SCOREBOARD reading '50 - 1 - 2.5 / STONES / KIDNEYS' — the numeric cheat-sheet for the surgical criteria: age <50, Ca >1 above ULN, T-score < -2.5, plus kidney findings.

WHAT IT ENCODES

The surgery-criteria mnemonic for asymptomatic PHPT: AGE <50; calcium >1.0 mg/dL above the upper limit of normal; T-score < -2.5 (or vertebral fracture); plus KIDNEY findings — eGFR <60, nephrolithiasis/nephrocalcinosis, or 24-hr urine calcium >400 mg/day. Any single one triggers operation. Memorize 'less than 50, one above, minus 2.5, and the kidneys.'

BOARD TRAP

A case lists a 24-hr urine calcium of 320 mg and asks if that alone justifies surgery — WRONG, that's below the 400 mg/day threshold. Conversely >400 mg/day with stone risk IS a criterion in many guideline versions. The discriminator: know the exact numbers (50 / 1 / 2.5 / GFR 60 / urine Ca 400); near-threshold values are deliberate distractors.

3. Any one trigger = operate

WHAT YOU SEE

Left banner ANY ONE TRIGGER = OPERATE — drives home the OR-logic: a single positive criterion sends the patient to surgery, no second one required.

WHAT IT ENCODES

The decision logic is OR, not AND: meeting any single guideline criterion is sufficient to indicate parathyroidectomy. A patient need not have all (or even two) criteria — one is enough. This reflects that even mild asymptomatic disease causes progressive bone loss, renal injury, and stone risk over time, all reversible/preventable with cure.

BOARD TRAP

A case where the patient meets only ONE criterion (e.g., age 45, otherwise normal) is dismissed as 'not enough indications, needs at least two to operate' — WRONG. A single criterion suffices. The discriminator: the rule is ANY-one, so do not require multiple positives before referring.

4. Age < 50

WHAT YOU SEE

An APPLE hanging on the tree labelled AGE < 50 (next to a young woman who gets a green check) — youth is its own surgical criterion because decades of high calcium lie ahead.

WHAT IT ENCODES

Age younger than 50 is an independent surgical criterion in asymptomatic PHPT. Younger patients have many decades of cumulative exposure to elevated calcium/PTH, so the lifetime risk of progressive bone loss, stones, and renal decline is high; cure now prevents these. In the scene the young woman (age 45) gets the green check — operate.

BOARD TRAP

A 45-year-old with mild asymptomatic PHPT and otherwise normal labs is offered 'monitor, she's too young and low-risk for surgery' — WRONG. Young age is itself an indication BECAUSE of the long exposure ahead, not a reason to defer. The discriminator: age <50 = operate, age >50 with no other criteria = can monitor.

5. Ca >1 mg/dL above ULN

WHAT YOU SEE

A blue GEM on the tree labelled 'Ca >1 mg/dL ABOVE ULN' — the precious-stone calcium criterion: only calcium more than 1.0 above the lab's upper limit counts, not any value above normal.

WHAT IT ENCODES

Serum calcium more than 1.0 mg/dL above the upper limit of normal (e.g., ULN 10.5 → criterion at >11.5 mg/dL) is a surgical indication. Use albumin-corrected or ionized calcium. The threshold is deliberately modest because even mildly elevated calcium in PHPT predicts progression; markedly high calcium also raises crisis risk.

BOARD TRAP

A patient with calcium 11.2 (ULN 10.5) is called 'meets the surgical calcium criterion' — WRONG; that's only 0.7 above ULN, under the >1.0 threshold. The discriminator: it is >1.0 mg/dL ABOVE the upper limit (not just any value above normal); compute the exact difference against that patient's lab ULN.

6. T-score < -2.5 (DEXA)

WHAT YOU SEE

A DEXA-scan icon on the tree reading 'T-SCORE < -2.5' — the bone-density criterion (any site, including the forearm) or a fragility/vertebral fracture sends the patient to surgery.

WHAT IT ENCODES

Osteoporosis by DEXA — a T-score < -2.5 at ANY site (lumbar spine, hip, or distal one-third radius) — OR a prior fragility/vertebral fracture (including silent fracture on imaging) is a surgical criterion. Because PHPT preferentially erodes CORTICAL bone, the distal radius is the most sensitive site and DEXA must include the forearm.

BOARD TRAP

A PHPT patient has normal spine and hip DEXA, the forearm is not scanned, and they are labeled 'no bone indication for surgery' — WRONG/incomplete. PHPT hits cortical bone (distal radius) first, so omitting the forearm can miss osteoporosis. The discriminator: always include the distal 1/3 radius on DEXA in PHPT; spine/hip alone can be falsely reassuring.

7. eGFR < 60

WHAT YOU SEE

A pair of KIDNEY icons on the tree labelled 'eGFR < 60' — the renal-function criterion: CKD stage 3 from PHPT is itself an operative trigger.

WHAT IT ENCODES

Renal impairment with eGFR <60 mL/min/1.73m² (CKD stage 3 or worse) attributable to PHPT is a surgical criterion. Chronic hypercalcemia and hypercalciuria cause nephrocalcinosis and tubulointerstitial damage; operating may halt further decline. eGFR <60 also limits some medical options (bisphosphonates) and raises secondary HPT confusion.

BOARD TRAP

A PHPT patient with eGFR 55 is told 'kidney function is fine, no renal indication' — WRONG; <60 is below the threshold and is itself an operative criterion. The discriminator: the renal cutoff is eGFR 60 (not 30 or 45); an eGFR in the 50s already qualifies.

8. Stones / nephrocalcinosis

WHAT YOU SEE

A KIDNEY studded with STONES on the tree labelled STONES / NEPHROCALCINOSIS — even a silent imaging-found stone (or 24-hr urine Ca >400) counts as a surgical criterion.

WHAT IT ENCODES

Nephrolithiasis or nephrocalcinosis (clinical OR detected on imaging), or 24-hr urine calcium >400 mg/day with increased stone risk, is a surgical indication. PHPT classically causes calcium-based stones via hypercalciuria. Even asymptomatic stones found incidentally on CT/ultrasound count — making renal imaging part of the PHPT workup.

BOARD TRAP

An asymptomatic PHPT patient with a 4 mm renal stone seen incidentally on CT is dismissed because 'he's never had renal colic' — WRONG. Silent stones or nephrocalcinosis on imaging count as a surgical criterion regardless of symptoms. The discriminator: imaging-detected stones qualify; absence of colic does not exclude the indication.

9. Parathyroidectomy = definitive (OR)

WHAT YOU SEE

A scrubbed SURGEON at left under an OR / PARATHYROIDECTOMY sign — the operating room is the only definitive cure, preferred whenever criteria are met and the patient is operable.

WHAT IT ENCODES

Parathyroidectomy is the ONLY definitive cure for PHPT and is preferred whenever criteria are met and the patient is operable. Preop localization (sestamibi scan, neck ultrasound, 4D-CT) guides minimally invasive single-gland resection; intraoperative PTH should drop >50% within 10 min (Miami criterion). Cure normalizes calcium, improves BMD, and reduces stone recurrence.

BOARD TRAP

A surgically eligible PHPT patient is offered 'lifelong cinacalcet' as equivalent definitive treatment — WRONG. Medical therapy only controls calcium or protects bone; it never cures. The discriminator: if criteria are met and the patient can tolerate surgery, parathyroidectomy is the answer; medical therapy is reserved for those who decline or are not operative candidates.

10. Cortical bone sites (radius/hip/spine)

WHAT YOU SEE

A bone diagram on the trunk pointing to DISTAL RADIUS / HIP / SPINE — flags that PHPT eats CORTICAL bone first (distal radius is the canary), so a normal spine can be falsely reassuring.

WHAT IT ENCODES

PHPT (excess PTH) preferentially resorbs CORTICAL bone, with the distal third of the radius the earliest and most affected site; trabecular spine is relatively spared early. This is why DEXA in PHPT must assess all three sites including the forearm. Severe long-standing disease causes osteitis fibrosa cystica (subperiosteal resorption, 'salt-and-pepper' skull, brown tumors).

BOARD TRAP

A PHPT patient with a normal spine T-score is reassured their bones are protected — WRONG, because PHPT spares trabecular spine relatively while eroding cortical radius. The discriminator: in PHPT the radius (cortical) is the canary, not the spine (trabecular); a normal spine does not rule out PHPT bone disease.

11. No criteria = monitor

WHAT YOU SEE

Right banner NO CRITERIA = MONITOR over an OBSERVATION LOUNGE where an older man (Ca only 0.3 above ULN, no stones) relaxes — criterion-negative PHPT goes to surveillance, not the OR.

WHAT IT ENCODES

If a patient with asymptomatic PHPT meets NO surgical criteria (age >50, Ca <1 above ULN, T-score > -2.5, no fracture, eGFR >60, no stones/normal urine Ca), active surveillance is appropriate. The older man in the scene with calcium 0.3 above ULN, T-score > -2.5, eGFR ok, and no stones gets monitoring rather than the OR.

BOARD TRAP

An asymptomatic PHPT patient who meets no criteria is pushed to surgery 'just to be safe' — WRONG/overtreatment. With no criteria met, guideline-concordant care is monitoring, not operating. The discriminator: surgery requires a met criterion (or symptoms); a fully criterion-negative patient is observed.

12. Serial monitoring panel

WHAT YOU SEE

An observation-lounge sign reading SERIAL MONITORING — serum Ca, renal function, DEXA, stone assessment — the watch-list for non-operative PHPT, because density can fall past -2.5 even while calcium stays flat.

WHAT IT ENCODES

Surveillance of non-operative PHPT includes serum calcium annually, serum creatinine/eGFR annually, and DEXA (3 sites including forearm) every 1-2 years. Crossing a surgical threshold during follow-up converts the patient to a surgical candidate. Renal imaging is checked if stone symptoms or rising urine calcium arise.

BOARD TRAP

A monitored PHPT patient skips DEXA for years because calcium has been 'stable' — WRONG. Calcium can stay flat while bone density silently declines past -2.5, which is itself an operative trigger. The discriminator: monitoring is multi-parameter (Ca + renal + DEXA); stable calcium alone does not confirm stable disease.

13. Medical options (cinacalcet / bisphosphonate)

WHAT YOU SEE

Bottom-left signpost MEDICAL OPTIONS: 'bisphosphonate = BMD not Ca', 'cinacalcet = Ca down', 'hydration supportive' — the drug menu for non-surgical patients, matching each agent to its single job.

WHAT IT ENCODES

When surgery is declined or contraindicated: cinacalcet (calcimimetic, activates CaSR) LOWERS serum calcium but does little for bone density; bisphosphonates (e.g., alendronate) improve/protect BMD but do NOT durably lower calcium. They address different problems, so a patient needing both effects may take both. Maintain hydration; ensure adequate (not excessive) vitamin D.

BOARD TRAP

A non-operative PHPT patient with hypercalcemia is started on a bisphosphonate to bring the calcium down — WRONG. Bisphosphonates raise BMD but do not correct hypercalcemia; cinacalcet is the agent that lowers calcium. The discriminator: match the drug to the goal — calcium → cinacalcet, bone → bisphosphonate.

14. Avoid thiazides / Li (monitor side)

WHAT YOU SEE

Right panel listing HYDRATION plus crossed-out THIAZIDES / Li — flags the two drugs that themselves raise calcium and can masquerade as PHPT, so stop them and recheck before diagnosing.

WHAT IT ENCODES

In PHPT, maintain hydration and AVOID provocateurs of hypercalcemia: thiazide diuretics (increase distal tubular calcium reabsorption → raise serum calcium and can unmask PHPT) and lithium (shifts the parathyroid set-point, raising PTH and calcium). If a patient on lithium/thiazide has borderline hypercalcemia, stop the drug and recheck before diagnosing PHPT.

BOARD TRAP

A patient on a thiazide has calcium 10.8 and is immediately diagnosed with PHPT — WRONG/premature. Thiazides (and lithium) themselves raise calcium; you must stop the offending drug and recheck calcium and PTH before committing to a PHPT diagnosis. The discriminator: drug-induced hypercalcemia mimics PHPT — clear the meds first.